

Research Articles

Contrast-Level Aggregate-Data Dose–Response Meta-Analysis for Heterogeneous Exposure Intervals

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Abstract

Meta-analysis is complicated when studies report adjusted odds ratios for study-specific pairs of exposure or intervention intervals but individual data, group frequencies, and outcome counts are unavailable. We propose a support-based contrast-level aggregate-data dose–response meta-analysis method for interval and exact supports. An outcome-independent support-to-design operator maps each comparison and reference support to the expected dose–response basis under a working exposure distribution. With the implemented linear basis, this is the support-specific mean; an exact value is a zero-width interval mapped to itself. Differences between the mapped supports are used as design covariates in a contrast-level working mixed model fitted by generalized least squares or profile restricted maximum likelihood. The approach retains arbitrary support pairs without recoding them to a common reference interval. Conventional point-dose and midpoint dose–response meta-analysis are exact special cases.

Under a correctly specified Gamma working distribution, slope bias was less than 0.001 in the empirical mixed-support simulation; severe distributional misspecification produced bias of -0.290 . In the principal 16-study residential-radon illustration, a historical country-specific Lognormal mapping gave an odds ratio of 1.081 per 100 Bq/m³ (95% confidence interval 1.032–1.133), with sensitivity to the assumed within-study correlation. A motivating eight-study animal-feeding-operation working analysis combines explicit-distance odds ratios and proxy mappings, with OR-only and explicit-support restriction analyses. Published randomized-trial meta-analyses provide examples of the same support-representation problem. The method is appropriate when the working exposure distribution is defensible; otherwise sensitivity analysis is essential.

Highlights

What is already known

Conventional dose–response meta-analysis generally assigns a single exposure value to each reported exposure category. Existing methods for grouped exposures often require individual participant data or group-level exposure–outcome information that may not be available in published studies.

What is new

We introduce a support-to-design operator for contrast-level aggregate data when published studies provide pairwise adjusted odds ratios and their uncertainty but the comparison and reference exposures are heterogeneous intervals or exact points. The operator maps each support to a distribution-dependent average dose–response basis vector before standard meta-regression. It requires neither exposure-group frequencies nor outcome counts, retains arbitrary reported support pairs, and contains conventional point-dose and midpoint dose–response meta-analysis as exact special cases.

Potential impact for Research Synthesis Methods readers

The method permits dose–response synthesis when comparison and reference intervals differ across studies and only reported contrasts and their uncertainty measures are available. It also accommodates mixtures of positive- and zero-width intervals and makes the working exposure-distribution assumption explicit. It therefore replaces an otherwise analyst-chosen point score with an auditable support model rather than eliminating the need for a within-support assumption. The support problem can arise in observational exposure studies and in randomized trials that assign intervention target ranges.

1. Introduction

Synthesizing research results from multiple studies is essential for understanding patterns of risk and informing policy. The use of meta-analysis for the quantitative synthesis of randomized controlled trials (RCTs) and observational studies is well established. A recurring difficulty arises when continuous exposure or intervention levels are reported through heterogeneous intervals rather than common point values. This problem is often encountered in observational research. When assessing unintended and unplanned exposures that may cause disease, observational studies are often necessary for ethical and practical reasons. Because such exposures are not deliberately assigned or standardized, the exposure definitions and effect measures reported by observational studies may vary considerably, making quantitative synthesis difficult. An example of this problem in environmental health is the association between exposure to emissions from large animal production facilities and the health of residents in surrounding communities [1, 2](#). This topic has been of interest to public health and environmental researchers for many years and has generated a body of evidence with disparate exposure measures, outcomes, and effect measures. This variation has made quantitative synthesis difficult. In this study, we address several of the challenges involved in synthesizing this body of evidence.

Conventional dose–response meta-analysis (DRMA) based on contrast-level aggregate data generally requires each exposure category to be represented by a single quantitative exposure value before a dose–response trend can be estimated [3, 4](#). A practical difficulty arises when studies report exposure through intervals with different boundaries and only the resulting reported association estimates and confidence intervals are available. In this setting, the original individual-level exposures, exposure-group counts, and observed mean exposures within the reported categories may be unavailable.

Related methods have addressed grouped or interval-censored exposures, but they require information that is not typically available when a meta-analysis is based on published effect estimates. Cook et al. [5](#) developed a retrospective likelihood approach using the joint information on exposure categories and disease status. That likelihood cannot be constructed when the primary studies report only odds ratios and confidence intervals without the underlying individual-level data or grouped exposure–outcome counts.

Allen et al. 6, 7 proposed estimating the mean exposure within reported intervals as a dose pre-analysis step. Their procedure uses the interval boundaries together with the numbers or proportions of individuals in the exposure groups to estimate the exposure distribution. It then uses reconstructed effective counts in a Bayesian dose–response likelihood. Such exposure-group frequencies are not consistently available in the studies included in the present review. Moreover, their grouped-data likelihood is not applicable to reported adjusted relative association estimates comparing different pairs of exposure intervals. Their applied pipeline is useful when those richer grouped data are available, but it is not applicable when the input consists only of adjusted relative association estimates, confidence intervals, and support bounds. Calculation of a conditional group mean is therefore not itself claimed as new here. Our contribution is a general support-level dose–response model, its formal properties, and inference for contrast-level summary data without group frequencies or exposure–outcome counts.

Takahashi and Tango 8 considered the assignment of representative exposure values when summarized response data are available for grouped exposure categories. Their likelihood requires category-specific response information that is not recoverable from reported odds ratios and confidence intervals alone.

Shi and Copas 9 modeled arbitrarily aggregated dose levels, and Hartemink et al. 10 addressed differing exposure cutpoints and nonlinear dose–response functions. Both are important close precedents, but their formulations use category-level structures and ordinarily a study reference category rather than arbitrary pairs of published adjusted contrasts. Riley et al. 11 developed multivariate meta-analysis for prognostic effects reported at multiple cutpoints, including within-study dependence and functional trends across cutpoints. That framework synthesizes cutpoint-specific prognostic effects rather than mapping arbitrary pairs of exposure supports to representative exposure contrasts. Yu et al. 12 compared conventional one- and two-stage summarized-data DRMA using all reported categories. Most recently, Bauleo et al. 13 proposed a categorical-to-continuous conversion based on category midpoints, category-specific continuous coefficients, and independence of those estimates. That practical procedure therefore retains the midpoint and covariance assumptions examined here.

The present review therefore has a different and common contrast-level aggregate-data structure. Here, contrast-level aggregate data refer to reported adjusted OR estimates and confidence intervals for comparisons between exposure groups, in contrast to group-level aggregate data such as event counts and denominators within each exposure interval. Each available observation is a reported adjusted OR estimate comparing a study-specific pair of exposure intervals, while the underlying individual participant data and group-level aggregate data are unavailable. The comparison and reference intervals may differ across observations, and multiple observations from the same study may involve overlapping or nested intervals. Exact exposure values are represented as intervals of width zero. Consequently, no method we identified is applicable to the complete combination of reported inputs in this meta-analysis: adjusted contrasts whose comparison and reference supports vary by observation, mixtures of intervals and exact values, within-study dependence, and no exposure-group frequencies or exposure–outcome counts. The close methods above remain complementary when their richer or more restrictive input structures are available.

Conventional summarized-data dose–response meta-analysis becomes operable only after the analyst supplies a point score for every support, while retrospective and dose pre-analysis methods require grouped exposure frequencies or exposure–outcome information that is missing here. The unresolved problem is therefore not the absence of a downstream meta-regression estimator, but the absence of a principled mapping from each reported pair of heterogeneous supports to the quantitative contrast required by that estimator. We formalize that mapping as the support-to-design operator

$$\mathcal{M}_{F_k, h}(S) = E_{F_k} \{h(D) \mid D \in S\}, \quad X_{kj} = \mathcal{M}_{F_k, h}(A_{kj}) - \mathcal{M}_{F_k, h}(B_{kj}), \quad (1.1)$$

Here, D is the continuous exposure, F_k is the working exposure distribution for study k , h is the selected dose–response basis, S is an exposure support, and A_{kj} and B_{kj} are the comparison and reference supports for observation j in study k . The operator separates outcome-independent support harmonization from conventional second-stage synthesis. This also separates *support uncertainty* in the first-stage mapping from *synthesis uncertainty* conditional on that mapping.

The same support-representation problem also occurs in randomized trials, which may assign participants to target ranges rather than exact doses, leaving the same support-representation problem when a meta-analysis uses published treatment contrasts. Two published examples motivate this broader application. Zhou et al. 14 synthesized 24 randomized trials involving 10,361 patients with chronic kidney disease whose higher and lower hemoglobin target supports varied across trials. Cumpstey et al. 15 synthesized eight randomized trials involving 4,415 mechanically ventilated patients and explicitly identified heterogeneous and overlapping oxygen target ranges. These examples show that interval-valued supports also arise in clinical-trial meta-analysis; implementation of the proposed dose–response model requires variation in the target contrasts and a justified working distribution within each target range.

Our approach addresses this setting by representing each comparison and reference interval by the mean exposure within that interval under a specified working exposure distribution. An exact value is represented as a zero-width interval whose representative mean is the point itself. Their difference defines the quantitative exposure contrast associated with the reported log odds ratio, and the resulting contrasts are synthesized using a random-effects meta-regression.

We obtained data from an ongoing living systematic review that builds on two previously published systematic reviews of associations between residential exposure to animal feeding operations and human health 1, 2, 16, 17. The studies identified by the review reported five major categories of health outcomes: lower respiratory diseases, upper respiratory diseases, antimicrobial resistance, infectious conditions, and gastrointestinal diseases.

The present analysis focused on lower respiratory disease (LRD) outcomes and distance-based measures of residential proximity to the nearest animal feeding operation (AFO), because these formed the largest subset of evidence that could be represented using distance intervals. Across studies, residential proximity was reported using heterogeneous categories and threshold-based contrasts, creating a need for a method that maps the reported comparisons onto a common distance scale.

Our objective was to develop and evaluate a contrast-level aggregate-data dose–response meta-analysis method for heterogeneous exposure intervals. The method represents each reported positive-width exposure interval by its mean exposure under a specified working exposure distribution, while an exact exposure value is represented by the point itself. The difference between the mean exposures in the comparison and reference supports defines a quantitative exposure contrast that can be incorporated directly into a random-effects meta-regression of the reported log odds-ratio estimates.

The proposed model is analogous to a conventional fixed-dose dose–response model: the difference between two fixed exposure values is replaced by the difference between the mean exposures within two reported intervals. When every support has width zero, this replacement is the identity and the proposed model is exactly the conventional point-dose model. It therefore inherits the usual linear or nonlinear dose–response basis, fixed- or random-effects structure, GLS or REML estimation, covariance modeling, and diagnostics in that special case. It accommodates heterogeneous interval boundaries and reference groups without requiring individual participant data or group-level aggregate data, and without recoding all comparisons to a common reference category. The generalized support family additionally includes exact points. The usual midpoint assignment is recovered as a special case when exposure is uniformly distributed within a bounded interval. Although motivated by residential distance, the same contrast-level aggregate-data structure can arise for other continuous exposures reported through study-specific intervals.

2. Method

2.1. Contrast-level aggregate-data structure with heterogeneous exposure intervals

We consider a set of K studies, indexed by $k = 1, \dots, K$, where study k reports n_k adjusted log odds ratio (log OR) estimates comparing outcomes between exposure-defined groups. The number of comparisons may vary across studies, reflecting differences in exposure categorization and study design. Let $y_{kj} = \log OR_{kj}$ denote this observed estimate for comparison j in study k . The statistical development below is formulated for log ORs. Any use of another reported measure as an approximate OR is an application-specific working assumption that must be justified and examined in sensitivity analysis.

Let A_{kj} and B_{kj} denote the comparison and reference exposure intervals, respectively, for observation j in study k . The corresponding contrast-level aggregate-data observation is the reported adjusted log OR for this support pair,

$$y_{kj} = \log \widehat{OR}_{kj}^{\text{adj}}(A_{kj} : B_{kj}).$$

The study-specific covariate adjustment or standardization is part of the reported estimand and cannot be reconstructed from contrast-level aggregate data alone.

The comparison and reference intervals may differ across observations and across studies. Multiple observations from the same study may also involve overlapping or nested intervals rather than a common set of mutually exclusive exposure categories.

All practical supports are represented by the unified interval notation

$$I(a, b) = \begin{cases} (a, b], & a < b, \\ \{a\} = [a, a], & a = b, \end{cases} \quad 0 \leq a \leq b < \infty. \quad (2.1)$$

Thus, an exact exposure value is the special case of an interval with width zero. In practice, even a category reported as $D > d$ arises from a study population with finite operational exposure support and can be represented as $(d, d_{\max}]$, where $d_{\max}$ is the study-specific maximum distance used to delimit that population. This was the convention used in the empirical illustration. A genuinely right-unbounded support (d, ∞) is also accommodated as a theoretical special case, as described below.

Each reported OR is accompanied by a 95% confidence interval. Under standard large-sample approximations, the log OR is approximately normally distributed [3](#), [4](#). The corresponding standard error is recovered from the reported confidence limits as:

$$\hat{\sigma}_{kj} = \frac{\log(UL_{kj}) - \log(LL_{kj})}{2 \times 1.96}, \quad (2.2)$$

and is used as the estimated sampling standard deviation of the observed log OR under the normal approximation. Here, UL_{kj} and LL_{kj} denote the upper and lower bounds of the reported 95% confidence interval for comparison j in study k . These standard errors define the diagonal elements of the within-study covariance matrix Σ_k .

Thus, the required aggregate data for each observation are the reported effect estimate, its standard error or confidence interval, and the boundaries of the comparison and reference exposure intervals. For a zero-width interval, the coincident lower and upper boundaries identify the exact point value. The method additionally requires a working exposure distribution from which the mean exposure within each positive-width interval can be calculated. Individual participant data, exposure-group frequencies, and grouped exposure–outcome counts are not required.

The exposure distribution is a first-stage, outcome-independent component of the analysis and is not nonparametrically identified by the reported relative association estimates alone without additional

restrictions on both the exposure family and the outcome model. We therefore recommend the following hierarchy. First, use a reported empirical group mean, or a study-population exposure distribution, when either is available. Second, when only exposure-interval frequencies are reported, fit a scientifically plausible distribution by the corresponding grouped-data likelihood and calculate its conditional means; a Lognormal model for multiplicative environmental exposures is one example 6. Third, use a population-relevant external distribution, with explicit justification and sensitivity analysis. When none of these sources is available, midpoint assignment remains an identifiable Uniform working-model special case, but it should be labeled as such and accompanied by sensitivity analyses over plausible skewed distributions and support bounds. Distribution selection should not use the outcome contrasts, because doing so would make the exposure scores outcome-adaptive. If several families remain plausible, the meta-analysis can be repeated across them or the support means can be averaged using prespecified first-stage model weights. Uncertainty from an estimated first-stage distribution can be propagated as described in Section 2.5.

Practical algorithm. For application, the support-to-design procedure has five steps:

1. retain each adjusted log OR, its standard error, study identifier, and the reported comparison and reference supports;
2. select an outcome-independent exposure distribution using the hierarchy above and record the evidence supporting that choice;
3. calculate the conditional mean of every positive-width support and use the reported value for every zero-width support;
4. subtract the reference mean from the comparison mean for each contrast and fit the GLS or profile-REML model, preserving within-study blocks; and
5. report sensitivity to plausible exposure distributions, support bounds, working correlations, and finite-study inference. If no distribution can be supported, report midpoint scoring explicitly as a Uniform working-model analysis rather than as an observed mean.

Worked support-to-design example. In the radon illustration, Blot et al.'s adjusted OR of 0.90 compares 74–147 Bq/m³ with the reported reference support below 74 Bq/m³. Under the prespecified Chinese Lognormal distribution (geometric mean 20, geometric standard deviation 2.2), the representative-mean definition in Equation 2.5 gives conditional means 96.47 and 23.18 Bq/m³. The resulting design value is $x = 96.47 - 23.18 = 73.30$ Bq/m³, and the response supplied to the second-stage model is $y = \log(0.90) = -0.105$. This calculation is repeated row by row; an exact exposure would enter directly as its reported point value. The analysis-ready workbook records the reported bounds, distribution parameters, conditional means, and design contrasts so the complete mapping is auditable.

The observed log ORs reported within the same study are likely to be correlated because they are derived from the same underlying study population and may involve overlapping exposure groups. For example, a study may report one comparison between $(0, 3]$ and $(3, d_{\max}]$ and another between $(0, 6]$ and $(6, d_{\max}]$. Residents contributing to the first comparison may therefore also contribute to groups in the second comparison, inducing dependence among the reported estimates. Standard dose–response and multivariate meta-analysis methods account for such dependence through the within-study covariance structure 3, 4. However, the studies included in our analysis generally reported only odds ratios and confidence intervals, without sufficient information to reconstruct the full covariance matrix.

Therefore, we estimated the marginal variances of the log ORs using the reported confidence limits (Equation 2.2) and approximated the within-study covariance structure by assuming a common correlation parameter ρ across comparisons within the same study 18. The use of plausible working correlation assumptions when the within-study covariance structure is unavailable has been discussed in the multivariate meta-analysis literature 19, 20. This treats multiple estimates from a study in the same aggregate-data manner commonly used in multivariate meta-analysis when the exact covariance

matrix is unavailable. In the meta-analysis, we used $\rho = 0.5$ as the working correlation assumption and further evaluated the sensitivity of the results under alternative values of ρ .

Collecting all n_k log ORs within study k into a vector $\mathbf{y}_k$, the joint within-study distribution is specified as:

$$\mathbf{y}_k \sim \text{MVN}(\boldsymbol{\eta}_k, \boldsymbol{\Sigma}_k), \quad (2.3)$$

where $\boldsymbol{\Sigma}_k$ has diagonal entries $\hat{\sigma}_{kj}^2$ and off-diagonal entries $\rho \hat{\sigma}_{kj} \hat{\sigma}_{kj'}$, and $\boldsymbol{\eta}_k$ denotes the vector of model-implied log ORs defined in the subsequent section.

2.2. Support-based model formulation

Contrast-level working mean model Let y_{kj} denote the reported adjusted log OR comparing support A_{kj} with support B_{kj} . We specify the working mean directly at the level of the reported contrast:

$$E\{y_{kj} \mid A_{kj}, B_{kj}, F_k, b_{1k}\} = (\beta_1 + b_{1k})\{\mu_k(A_{kj}) - \mu_k(B_{kj})\}, \quad (2.4)$$

where:

- $\mu_k(S)$ is the representative mean distance for support S : the conditional mean under the working exposure distribution when $a < b$ and d when $a = b = d$,
- β_1 is the common dose–response slope, and
- $b_{1k} \sim N(0, \tau^2)$ is a study-specific random effect on the slope.

Estimand and working assumptions. The parameter β_1 is the slope in a support-level working model for the reported adjusted log OR contrasts. Without additional individual-level and causal assumptions, it is not an individual-level causal dose–response slope. The interpretation rests on the following explicit conditions.

1. The reported estimates are analyzed on a common log-OR scale. An effect measure used only as an application-specific proxy is identified and examined in a separate sensitivity analysis.
2. The exposure distribution F_k is chosen from outcome-independent information and is relevant to the source population. It is not estimated by selecting the distribution that best fits the reported outcome contrasts.
3. Conditional on the mapping and working covariance, the expected adjusted contrast is modeled as linear in the mapped support difference in Equation (2.4).
4. The point-dose and midpoint reductions are exact algebraic special cases of this working contrast design under the same basis and synthesis model; they do not make arbitrary adjusted category ORs exact aggregations of an individual-level logistic model.
5. Proposition 3 supplies a first-order individual-level justification for an unadjusted logistic model. With adjustment covariates, the corresponding first-order contrast may depend on covariate-conditional exposure distributions and outcome-risk weights.
6. Odds-ratio noncollapsibility, incompatible adjustment sets, residual confounding, and causal transportability are not resolved by the support mapping. These remain properties of the source estimates and their target populations.

The support mapping is identified from observed support means or a justified working distribution F_k ; without restrictive exposure and outcome models, it is not identified from the reported outcome contrasts alone.

Representative mean for an exposure support To evaluate the model-defined exposure contrasts, it is necessary to compute a representative mean distance for each exposure support. For the unified support

$\mathcal{I}(a, b)$, define

$$\mu_k\{\mathcal{I}(a, b)\} = \begin{cases} \frac{\int_a^b u f_k(u) du}{\int_a^b f_k(u) du}, & a < b, \\ d, & a = b = d. \end{cases} \quad (2.5)$$

Under the Gamma distribution for D defined in Section 2.7, the positive-width expectation can be obtained in closed form for any finite interval $(a, b]$. In the empirical illustration, categories described by a lower threshold were bounded by the corresponding study-specific maximum distance $d_{\max}$ to reflect the finite operational support of the contributing study population. Consequently, all interval calculations in the application have finite limits.

For a positive-width interval $(a, b]$, with $0 \leq a < b < \infty$, the conditional mean is

$$\mu_k(a, b) = E_k[D \mid a < D \leq b] = \alpha_k \theta_k \cdot \frac{P_{\alpha_k+1, \theta_k}(a < D \leq b)}{P_{\alpha_k, \theta_k}(a < D \leq b)},$$

where $P_{\alpha_k, \theta_k}(\cdot)$ denotes the $\text{Gamma}(\alpha_k, \theta_k)$ probability of the stated event under the working distribution used for study k . Thus, a category reported as $D > d$ is evaluated in the application as $\mu_k(d, d_{\max})$. If a genuinely right-unbounded support is required, the same theory gives the special case

$$\mu_k(d, \infty) = \alpha_k \theta_k \frac{P_{\alpha_k+1, \theta_k}(D > d)}{P_{\alpha_k, \theta_k}(D > d)},$$

provided the tail mean exists; it always exists under the Gamma working distribution. This expression can be evaluated from Gamma survival probabilities and does not require numerical integration to an infinite endpoint. For the zero-width case $a = b = d$, direct substitution into the ratio of integrals in Equation (2.5) gives $0/0$ under a continuous distribution. Its continuous extension is nevertheless the point distance: provided f_k is continuous and positive near d ,

$$\mu_k\{\mathcal{I}(d, d)\} = \lim_{\substack{a, b \rightarrow d; \\ 0 \leq a \leq d \leq b \\ a < b}} \frac{\int_a^b u f_k(u) du}{\int_a^b f_k(u) du} = d. \quad (2.6)$$

Thus, no numerical integration is performed for a zero-width interval; its representative mean is evaluated directly as d .

Proposition 1 (point-dose and midpoint DRMA as nested special cases). Let $\mathbf{h}(d)$ be a p -dimensional vector of dose-response basis functions and define

$$\mathbf{m}_k(S) = \mathcal{M}_{F_k, h}(S) = E_k\{\mathbf{h}(D) \mid D \in S\}, \quad \mathbf{m}_k\{\mathcal{I}(d, d)\} = \mathbf{h}(d), \quad (2.7)$$

where the point value is again the shrinking-interval extension when the basis functions are continuous at d . If a latent dose-response function on the chosen linear-predictor scale is $\eta_k(d) = \gamma_k + \boldsymbol{\beta}_k^\top \mathbf{h}(d)$, linearity of conditional expectation gives the exact support-averaging identity

$$E_k\{\eta_k(D) \mid D \in S\} = \gamma_k + \boldsymbol{\beta}_k^\top \mathbf{m}_k(S). \quad (2.8)$$

Consequently, the contrast between supports A and B has design vector $\mathbf{m}_k(A) - \mathbf{m}_k(B)$, with the following two nested special cases.

(i) *Point-dose DRMA.* If all supports have width zero, so that $A_{kj} = \{d_{kj,A}\}$ and $B_{kj} = \{d_{kj,B}\}$, then $\mathbf{m}_k(A_{kj}) - \mathbf{m}_k(B_{kj}) = \mathbf{h}(d_{kj,A}) - \mathbf{h}(d_{kj,B})$. The model, sampling covariance, fixed- or random-effects likelihood, GLS or REML estimator, standard errors, and diagnostics are exactly those of conventional

point-dose contrast-design dose-response meta-analysis using the same basis, sampling covariance, and random-effects specification.

(ii) *Midpoint DRMA*. For the present linear model, $p = 1$ and $h(d) = d$. If, conditional on every bounded positive-width support $S = \mathcal{I}(a, b)$, exposure is Uniform on $(a, b]$, then $m_k(S) = E_k(D \mid a < D \leq b) = (a + b)/2$. Hence each support contrast is exactly the difference between the assigned category midpoints, and the complete model reduces to conventional midpoint-scored DRMA. This result concerns the conditional distribution within each reported support; the category probabilities need not be equal.

Positive-width and zero-width supports can be mixed because the mapping is applied separately to every reported comparison. For nonlinear bases such as regression splines, the corresponding extension averages each basis function over the support, rather than evaluating the basis only at the interval midpoint. This identity concerns the support-averaged linear predictor; with nonlinear outcome links and no grouped outcome counts, the reported marginal log OR relationship remains a contrast-level working model as stated above. The public interface and applications in this paper implement and evaluate the linear support-mean model; a general nonlinear basis implementation is a future extension. The accompanying validation suite also checks this reduction numerically: an all-point dataset fitted through the generalized interface reproduces a separately assembled conventional GLS/profile-REML calculation to machine precision.

The proof is given in Supplementary Methods S4.

The conditional mean is also the unique minimum-MSE constant representation of D within a support under squared-error loss, and it is coherent when supports are split or combined using their exposure probabilities. This elementary projection property motivates the representative value but is not a claim of minimum MSE for $\hat{\beta}_1$ or of optimality under arbitrary outcome links; the full statement and proof are in Supplementary Methods S4.

This formulation represents each interval by its mean distance under the specified working exposure distribution and generalizes the midpoint assignments used in traditional dose-response meta-analysis. Exact-distance comparisons are represented by their point values, so ordinary fixed-dose contrasts are included as a special case.

Contrast design and observed-data model Define the study-level covariate:

$$x_{kj} = \mu_k(A_{kj}) - \mu_k(B_{kj}). \quad (2.9)$$

This leads to the working model:

$$y_{kj} = (\beta_1 + b_{1k})x_{kj} + \varepsilon_{kj}, \quad (2.10)$$

where ε_{kj} represents sampling error with covariance structure Σ_k , and b_{1k} is a study-specific deviation from the common slope β_1 .

Accordingly, the reported adjusted log OR estimates have the linear mixed-effects representation in Equation (2.10), with the exposure contrast x_{kj} defined in Equation (2.9). Once each reported comparison is mapped to the difference between its support-specific representative mean distances, comparisons involving heterogeneous exposure supports can be analyzed within a common meta-analytic regression framework.

The model includes a study-specific term b_{1k} to allow for variation in the dose-response relationship across studies. Such variation is expected given differences in study settings and designs among the included studies.

Depending on how this heterogeneity is treated, two model specifications are considered: a fixed-effect model, which assumes a common slope across studies ($b_{1k} = 0$), and a random-effects model, which treats b_{1k} as a random variable. Given the diversity of the included studies, the random-effects model is taken as the primary specification, while the fixed-effect model is used as a reference.

Identifiability Conditional on the support-to-design mapping and a positive-definite working covariance, the common slope β_1 is identifiable if and only if the information $\mathcal{J}(\tau)$ in Proposition 2 is positive; equivalently, at least one design contrast is nonzero. This conditional statement does not imply that an unknown support mapping is jointly identified with the slope. If a low-dimensional parametric mapping $\mathbf{x}(\psi)$ is instead estimated from the outcome contrasts, a necessary local rank condition for the mean model is that

$$\left[\mathbf{x}(\psi), \beta_1 \frac{\partial \mathbf{x}(\psi)}{\partial \psi^\top} \right]$$

have full column rank. Multiple or overlapping supports can contribute such information only when changes in ψ alter the relative pattern of design contrasts. If $\mathbf{x}(\psi) = c(\psi)\mathbf{x}_0$, the change is only a common rescaling and is confounded with β_1 (and with τ in the random-effects model). Thus, overlapping supports do not by themselves identify an exposure distribution. An outcome-based profile may be useful as a model-compatibility diagnostic under explicit restrictions, but it is not an outcome-independent estimate of the source-population exposure distribution. The between-study variance parameter τ^2 is informed by variation in study-specific slopes and is imprecisely estimated when the number of independent studies is small.

2.3. Model estimation

Fixed-effect model The fixed-effect model assumes a single common dose–response slope β_1 shared across all K studies, with no between-study heterogeneity in the slope. This corresponds to setting $b_{1k} = 0$ for all k , so the working model in Equation (2.10) simplifies to:

$$y_{kj} = \beta_1 x_{kj} + \varepsilon_{kj}.$$

The marginal distribution of the observed log ORs within study k is:

$$\mathbf{y}_k \sim N(\beta_1 \mathbf{x}_k, \boldsymbol{\Sigma}_k). \quad (2.11)$$

This specification assumes that all studies share the same underlying distance–response relationship and does not account for between-study heterogeneity. Such an assumption seems strong, therefore this model serves primarily as a reference model for comparison with the random-effects specification.

Random-effects model We also consider a random-effects specification that allows the dose–response slope to vary between studies. Specifically, each study is assumed to have its own slope $\beta_1 + b_{1k}$, where b_{1k} represents a study-specific deviation from the common effect.

We assume:

$$b_{1k} \sim N(0, \tau^2), \quad \varepsilon_k \sim N(\mathbf{0}, \boldsymbol{\Sigma}_k),$$

independently. Marginally, this yields:

$$\mathbf{y}_k \sim N\left(\beta_1 \mathbf{x}_k, \boldsymbol{\Sigma}_k + \tau^2 \mathbf{x}_k \mathbf{x}_k^\top\right). \quad (2.12)$$

where τ^2 represents the between-study variance in the dose–response slope. A larger value of τ indicates greater variability across studies in the association between distance and disease risk. The additional covariance induced by the random slope has rank one, which reflects the fact that all comparisons within a study share the same study-specific deviation b_{1k} .

This specification accommodates differences in study settings and designs by allowing study-specific deviations from the common effect, while still pooling information across studies through β_1 .

Parameter interpretation The parameter β_1 represents the change in the model-implied log OR associated with a one-mile increase in the difference between the representative distances of the comparison and reference supports. A negative value of β_1 indicates that comparisons involving larger differences in mean residential distance are associated with lower odds of the reported LRD outcome. Because the underlying outcomes and study designs differ, β_1 should not be interpreted as a common individual-level risk or prevalence effect. It is a contrast-level aggregate-data estimand and should not be interpreted as the individual-level effect of moving one residence one mile farther from an AFO.

The parameter τ represents between-study heterogeneity in the distance–response slope. Larger values of τ indicate greater variation among studies in the support-level association between residential distance and the reported LRD outcome.

Marginal likelihood and REML estimation. Under the random-effects model, the marginal likelihood of the observed summary-level data is

$$L(\beta_1, \tau) = \prod_{k=1}^K \phi_{n_k} \left(\mathbf{y}_k; \beta_1 \mathbf{x}_k, \boldsymbol{\Sigma}_k + \tau^2 \mathbf{x}_k \mathbf{x}_k^\top \right),$$

where $\phi_{n_k}(\cdot; \mathbf{m}, V)$ denotes the n_k -dimensional multivariate normal density with mean vector $\mathbf{m}$ and covariance matrix V . The random-effects contribution to the covariance matrix enters through the rank-one term $\tau^2 \mathbf{x}_k \mathbf{x}_k^\top$. This structure preserves computational tractability even when multiple interval contrasts are reported within the same study.

The parameters (β_1, τ) are estimated using REML, which provides stable estimation of variance components in random-effects models. For a given value of τ , the generalized least squares (GLS) estimator of β_1 is:

$$\hat{\beta}_1(\tau) = \frac{\sum_{k=1}^K \mathbf{x}_k^\top V_k^{-1} \mathbf{y}_k}{\sum_{k=1}^K \mathbf{x}_k^\top V_k^{-1} \mathbf{x}_k}, \quad (2.13)$$

where:

$$V_k = \boldsymbol{\Sigma}_k + \tau^2 \mathbf{x}_k \mathbf{x}_k^\top.$$

For the fixed-effect model, $\tau = 0$ so $V_k = \boldsymbol{\Sigma}_k$, and Equation (2.13) reduces to the standard GLS estimator with standard error:

$$\widehat{\text{se}}(\hat{\beta}_1) = \left(\sum_{k=1}^K \mathbf{x}_k^\top \boldsymbol{\Sigma}_k^{-1} \mathbf{x}_k \right)^{-1/2}.$$

For the random-effects model, τ is estimated by profiling out $\beta_1(\tau)$ and maximizing the REML log-likelihood:

$$\ell_R(\tau) = -\frac{1}{2} \sum_{k=1}^K \left[\log |V_k| + (\mathbf{y}_k - \hat{\beta}_1 \mathbf{x}_k)^\top V_k^{-1} (\mathbf{y}_k - \hat{\beta}_1 \mathbf{x}_k) \right] - \frac{1}{2} \log \left(\sum_{k=1}^K \mathbf{x}_k^\top V_k^{-1} \mathbf{x}_k \right). \quad (2.14)$$

Because the restricted likelihood depends on the fixed-effect design, values of ℓ_R are used only to optimize τ within a fixed support mapping. They are not compared across the conditional-mean, midpoint, or alternative distribution designs; such a comparison would not be a valid model-selection test.

This formulation reduces estimation to a one-dimensional optimization over τ . In the released estimator, the design is divided by its root-mean-square value, optimization is performed on the resulting dimensionless scale, and $\hat{\beta}_1$, $\hat{\tau}$, and their uncertainty are transformed back to the original exposure units. The exact boundary $\tau = 0$ is compared with the interior optimum, the upper

bracket is expanded adaptively, and nonfinite criteria, optimizer failure, and boundary solutions are reported. A profile restricted-likelihood interval for τ is provided as an approximate descriptive interval; variance-component inference at $\tau = 0$ is nonregular.

The model-based confidence interval for β_1 uses the usual Normal critical value. Because both applications contain few independent study or dataset blocks, we additionally evaluated a finite-study sensitivity interval that replaces 1.96 by the 0.975 quantile of a Student t distribution with $K - 1$ degrees of freedom, without changing $\hat{\beta}_1$, $\hat{\tau}$, or the model-based standard error. This study-count correction is reported as a sensitivity analysis rather than as a full variance correction. The distinction is important because uncertainty in heterogeneity estimation can affect confidence-interval coverage when few studies are available [21, 22](#).

As a further model-based finite-study sensitivity analysis, we used a studentized parametric bootstrap conditional on the fitted support means and sampling covariance matrices. Each of 2,000 replicates drew one random slope per independent study or dataset block, drew the within-block multivariate sampling error, and refitted both β_1 and τ by profile REML. The confidence limits invert the 0.025 and 0.975 quantiles of the bootstrap statistic $(\hat{\beta}_1^{(b)} - \hat{\beta}_1)/\widehat{se}^{(b)}$. This analysis addresses second-stage finite-study uncertainty but remains conditional on the first-stage exposure representatives. We do not designate this interval as primary because its repeated-sampling coverage has not been established for the present small- K design.

For the 16-study radon illustration, we also calculated a CR2 cluster-sandwich variance sensitivity [23](#). At the fitted $\hat{\tau}$, a lower-triangular Cholesky factor L_k satisfying $L_k L_k^\top = \widehat{V}_k$ was used to form $\widetilde{\mathbf{y}}_k = L_k^{-1} \mathbf{y}_k$ and $\widetilde{\mathbf{x}}_k = L_k^{-1} \mathbf{x}_k$. Thus the CR2 target covariance after whitening was the identity matrix. With $\widetilde{\mathcal{J}} = \sum_k \widetilde{\mathbf{x}}_k \widetilde{\mathbf{x}}_k^\top$, $H_{kk} = \widetilde{\mathbf{x}}_k \widetilde{\mathbf{x}}_k^\top / \widetilde{\mathcal{J}}$, and $A_k = (I - H_{kk})^{-1/2}$ computed by symmetric eigendecomposition, the sensitivity variance was

$$\widehat{\text{Var}}_{\text{CR2}}(\widehat{\beta}_1) = \widetilde{\mathcal{J}}^{-2} \sum_k \left\{ \widetilde{\mathbf{x}}_k^\top A_k (\widetilde{\mathbf{y}}_k - \widehat{\beta}_1 \widetilde{\mathbf{x}}_k) \right\}^2.$$

Studies were the clustering units. The released NumPy implementation uses a t_{K-1} critical value; it does not calculate a Satterthwaite degrees-of-freedom approximation. We therefore describe this result as a CR2 variance estimate with a t_{K-1} critical value, not as a full CR2/Satterthwaite small-sample correction. With four studies, the corresponding AFO restriction was too small for this sensitivity to be informative.

Once $\widehat{\beta}_1$ and $\hat{\tau}$ are obtained, study-specific deviations b_{1k} are estimated using empirical best linear unbiased predictors (EBLUPs). Using the Woodbury matrix identity [24](#), the predictor can be written in the following numerically equivalent scalar form:

$$\widehat{b}_{1k} = \frac{\mathbf{x}_k^\top \Sigma_k^{-1} (\mathbf{y}_k - \widehat{\beta}_1 \mathbf{x}_k)}{\widehat{\tau}^{-2} + \mathbf{x}_k^\top \Sigma_k^{-1} \mathbf{x}_k}, \quad (2.15)$$

where $\widehat{b}_{1k}$ represents the estimated deviation of study k from the common slope, and $\widehat{\beta}_1 + \widehat{b}_{1k}$ is the corresponding study-specific slope estimate.

2.4. Relation to traditional dose–response meta-analysis

Classical dose–response meta-analysis models [3, 4](#) typically take the form:

$$\log OR_{kj} = (\beta_1 + b_{1k}) x_{kj},$$

where x_{kj} is usually defined as the difference between an assigned exposure value for category j and a study-specific reference category. Under this formulation, each reported effect is expressed relative to its study-specific reference category.

These summarized-data models use generalized least squares or related one- or two-stage methods to account for correlated estimates and between-study heterogeneity 25–27. The corresponding model-checking tools have also been described in this journal 26.

In the proposed approach, exposure contrasts are instead defined directly as $x_{kj} = \mu_k(A_{kj}) - \mu_k(B_{kj})$, representing the difference in mean residential distance between two intervals. This formulation directly accommodates each reported pair of comparison and reference intervals without recoding all comparisons to a common reference interval. The intervals may have positive or zero width.

Thus, the methodological addition is the support-to-design mapping upstream of the meta-regression, not a replacement for established GLS or REML machinery. Proposition 1 establishes that conventional point-dose DRMA is the all-zero-width special case. Consequently, every property of the conventional estimator that depends only on the resulting design matrix and covariance model carries over directly in that case. Positive-width intervals enlarge the admissible support family; points and intervals may occur together in one synthesis. Midpoint-based DRMA is obtained under the additional special-case assumption of a Uniform working distribution within every bounded interval.

The unresolved feature of the motivating evidence is not interval censoring alone, but the conjunction of four features: only reported contrasts and their standard errors are available; each contrast may compare its own pair of supports; positive-width intervals and exact values may occur together; and exposure-group frequencies or outcome counts are unavailable. The existing methods reviewed above address important subsets of this problem, but are not applicable to this combination of reported inputs alone 3–7. In usual practice, assigning a midpoint or endpoint can make an analysis possible, but that score is an approximation unless its implied within-support assumptions are appropriate. The conditional-mean projection result and Proposition 3 show, respectively, the excess within-support squared error and the additional first-order error generally introduced by midpoint assignment. The proposed mapping replaces that otherwise analyst-chosen step with an explicit working exposure model, permits its uncertainty to be propagated, and retains the reported pairwise contrast without reconstructing pseudo-arms or imposing a common reference support. The prior methods and the proposed method are therefore complementary; they target different available data structures.

Proposition 2 (conditional GLS identifiability and profile-REML unit equivariance). For specified τ and known support mappings, let $\mathbf{V}_k = \Sigma_k + \tau^2 \mathbf{x}_k \mathbf{x}_k^\top$ be positive definite and define

$$\mathcal{J}(\tau) = \sum_{k=1}^K \mathbf{x}_k^\top \mathbf{V}_k^{-1} \mathbf{x}_k. \quad (2.16)$$

The common slope is identifiable if and only if $\mathcal{J}(\tau) > 0$. Under the working model, the GLS estimator in Equation (2.13) is unbiased conditional on the support mappings and specified τ , with variance $\mathcal{J}(\tau)^{-1}$. This does not assert exact finite-sample unbiasedness of the plug-in estimator $\hat{\beta}_1(\hat{\tau})$; its performance is evaluated in the simulation study.

Moreover, under any positive affine change of exposure units $D^* = a + cD$, $c > 0$, the transformations $\mathbf{x}_k^* = c\mathbf{x}_k$, $\beta_1^* = \beta_1/c$, and $\tau^* = \tau/c$ leave every fitted log OR and marginal covariance matrix unchanged. The restricted log-likelihood satisfies $\ell_R^*(\tau/c) = \ell_R(\tau) - \log c$. Because the added term is constant in τ , $\hat{\tau}^* = \hat{\tau}/c$ and $\hat{\beta}_1^* = \hat{\beta}_1/c$; profile-likelihood differences and profile limits transform equivariantly.

The proof is given in Supplementary Methods S4. The public implementation standardizes the design internally, permits the exact boundary $\tau = 0$, and uses an adaptive dimensionless upper bracket before transforming $\hat{\beta}_1$, $\hat{\tau}$, and their uncertainty back to the reported exposure units. It also checks optimization status and reports boundary and convergence diagnostics.

To our knowledge, no method identified in our targeted search is applicable to the complete combination of already-adjusted arbitrary-pair contrasts, mixed point and interval supports, within-study dependence, and the absence of exposure-group frequencies or outcome counts. The contribution is therefore not simply replacing a midpoint: it is an outcome-independent support-to-design operator that makes this complete reported-data structure compatible with conventional synthesis. For example, the recent procedure of Bauleo et al. 13 uses category midpoints, category-specific per-unit coefficients, event-count weights, and an independence assumption; the present method retains arbitrary reported pairs and their working covariance without requiring category event counts. A full method comparison and the reproducible targeted-search strategy are reported in Supplementary Methods S3 and Supplementary Table S2. Under a Uniform working distribution, the conditional mean of a bounded interval equals its midpoint; under a nonuniform distribution, the two generally differ.

To evaluate how the proposed conditional-mean approach compares to the traditional interval-midpoint approach, we applied both approaches to our data and assessed them under the random-effects support-based model. Simulations reported bias and root mean squared error (RMSE) for $\hat{\beta}_1$ and $\hat{\tau}$, Normal and study-count- t interval coverage and mean width, type-I error under a null slope, REML convergence, and the proportion of fits at the exact $\tau = 0$ boundary. Monte Carlo standard errors were calculated for coverage and type-I error. The t_{K-1} interval is a transparent small-study sensitivity, not a complete variance correction. The studentized parametric- bootstrap intervals in the applications are additional model-based sensitivities because their repeated-sampling coverage was not validated here.

The main simulation grid and stress-test scenarios used ten independent study blocks. The main grid used a reproducible $K = 10$ template containing each of the eight empirical support-and-covariance block configurations once plus two resampled blocks; it therefore retained both positive-width and exact-point comparisons. The random-effects design crossed three multipliers (0.5, 1, and 2) of the empirical estimates $\hat{\beta}_1 = -0.069$ and $\hat{\tau} = 0.129$, producing a 3×3 grid. A 2×2 comparison additionally crossed two analysis approaches (midpoint vs. conditional mean) with two true exposure distributions (Uniform vs. Gamma). Under the Uniform setting, the two approaches are identical for positive-width intervals, and both evaluate zero-width intervals at their exact values. Under the Gamma setting, the conditional-mean approach uses the true representative interval means, whereas midpoint assignment ignores the skewed within-interval distribution. Each setting used $R = 5,000$ replicates.

Study-count generalizability. To evaluate generalizability across and beyond the observed number of studies, a separate experiment used $K = 10, 20$, and 30 independent study blocks. For each K , 25 support-and-covariance templates were formed by sampling the empirical study-block configurations with replacement. Data were generated at the empirical slope and heterogeneity, with 5,000 total replicates per cell. This design varies the number and mixture of support configurations without assuming that one arbitrarily selected subset of studies represents each K .

Additional stress tests evaluated point-estimation performance when a Lognormal or Uniform exposure distribution generated the interval means but a Gamma distribution was fitted, when the true $d_{\max}$ values were 20% lower or higher than the fitted values, and when the true within-study correlation was 0.3 or 0.7 but $\rho = 0.5$ was fitted. To represent uncertainty in the externally estimated exposure distributions, each regional Gamma shape and scale parameter was independently perturbed in each replicate using mean-preserving Lognormal draws with coefficients of variation of 10% or 20%; data were generated under the corresponding unperturbed regional Gamma parameters. These values define scenario-based perturbations because standard errors were not available for every external regional fit.

To assess the support-level working approximation, further datasets were generated from an individual-level logistic model with baseline log odds -2.2 , corresponding to a baseline disease probability of $\text{expit}(-2.2) = 0.100$; this value was chosen as a representative 10% probability rather than estimated from either application. For a positive-width support, the disease probability was obtained by integrating the individual probabilities over the true conditional Gamma distribution; for a zero-width support it was evaluated at the exact point. Logit-linear slopes of 0.5, 1, and 2 times the empirical $\hat{\beta}_1$

were examined. Nonlinear settings used $-2.2 + s_k d + c \log(1 + d)$ with $s_k = \beta_1 + b_{1k}$ and curvature $c = 0.05$ or 0.10 , while the fitted support-level model remained linear in representative distance. All stress tests retained the $K = 10$ mixed-support template and were summarized using bias and RMSE.

The following result is the main individual-level approximation justification for the conditional-mean mapping; the minimum-MSE projection property is a support-representation result rather than proof of universal optimality for the dose–response estimator.

Proposition 3 (first-order comparison with midpoint assignment). To further examine the relationship between the proposed interval-level model and an individual-level dose–response model, suppose that the exact exposure satisfies

$$\text{logit}\{P(Y = 1 \mid D = d, \text{ study } k)\} = \gamma_k + s_k d,$$

where $s_k = \beta_1 + b_{1k}$. For the fixed-effect model, $s_k = \beta_1$. Let

$$\mu_k(A) = E_k(D \mid D \in A), \quad v_k(A) = \text{Var}_k(D \mid D \in A),$$

and let $p_{0k} = \text{expit}(\gamma_k)$. The actual log odds after aggregating the individual probabilities within interval A is

$$L_k(A) = \text{logit} [E_k\{\text{expit}(\gamma_k + s_k D) \mid D \in A\}].$$

For a fixed interval with a finite third conditional moment, a Taylor expansion as $s_k \rightarrow 0$ gives

$$L_k(A) = \gamma_k + s_k \mu_k(A) + \frac{s_k^2}{2} (1 - 2p_{0k}) v_k(A) + O(s_k^3).$$

Consequently, for comparison and reference intervals A and B ,

$$\begin{aligned} L_k(A) - L_k(B) &= s_k \{\mu_k(A) - \mu_k(B)\} \\ &\quad + \frac{s_k^2}{2} (1 - 2p_{0k}) \{v_k(A) - v_k(B)\} + O(s_k^3). \end{aligned}$$

Thus, the interval-mean contrast used in the proposed model recovers the first-order term of the aggregate log odds ratio under the individual-level logit-linear model. In contrast, if $m(A)$ and $m(B)$ denote the interval midpoints, midpoint assignment introduces the additional first-order error

$$s_k [\{\mu_k(A) - m(A)\} - \{\mu_k(B) - m(B)\}].$$

Therefore, the error of the interval-mean approximation begins at order s_k^2 , whereas midpoint assignment generally has an additional order- s_k error unless the midpoint errors for the two intervals cancel. The two approaches coincide when the interval means equal their midpoints, including when exposure is uniformly distributed within both bounded intervals. For a zero-width interval, the conditional variance is zero and the support-level expression is exact: $L_k\{I(d, d)\} = \gamma_k + s_k d$. $\square$

The expansion is with respect to the dose–response slope rather than the interval width and therefore does not require narrow exposure intervals. However, the magnitude of the higher-order terms depends on both the slope and the conditional spread and higher moments of exposure within the intervals. Accordingly, the approximation may be less accurate for strong dose–response relationships or very wide or heavy-tailed exposure intervals. This proposition concerns an unadjusted individual-level logistic model. It is not a derivation for arbitrary adjusted ORs, whose first-order aggregation can depend on covariate-conditional exposure distributions, outcome-risk weights, and the primary study’s adjustment or standardization procedure.

2.5. Support-distribution uncertainty and model checking

The likelihood above is conditional on the representative support values. In applications such as the AFO illustration, these values may depend on an estimated first-stage parameter vector ψ , such as regional Gamma shape and scale parameters. Write $\hat{\boldsymbol{\vartheta}}(\psi)$ for the fitted meta-analytic parameters, where $\boldsymbol{\vartheta} = (\beta_1, \tau)^\top$, and let $\hat{\mathbf{C}}_{\text{cond}}$ denote their covariance matrix conditional on ψ .

Proposition 4 (first-stage uncertainty propagation). Suppose that the first-stage estimator $\hat{\psi}$ is independent of the reported meta-analytic contrasts, is asymptotically normal with estimated covariance $\hat{\mathbf{C}}_\psi$, and that $\hat{\boldsymbol{\vartheta}}(\psi)$ is differentiable in a neighborhood of the fitted value, with $\tau > 0$ an interior variance-component value. Then a first-order unconditional covariance estimator is

$$\hat{\mathbf{C}}_{\text{total}} = \hat{\mathbf{C}}_{\text{cond}} + \hat{\mathbf{G}}\hat{\mathbf{C}}_\psi\hat{\mathbf{G}}^\top, \quad \hat{\mathbf{G}} = \left. \frac{\partial \hat{\boldsymbol{\vartheta}}(\psi)}{\partial \psi^\top} \right|_{\hat{\psi}}. \quad (2.17)$$

The derivative can be evaluated analytically or by refitting the model after small numerical perturbations of the support-distribution parameters. Rows whose comparison and reference supports are both exact points have zero derivative with respect to ψ and therefore add no first-stage exposure-distribution uncertainty.

The proof is given in Supplementary Methods S4.

A parametric bootstrap provides a corresponding simulation-based procedure. One draws $\psi^{(b)}$ from its estimated sampling distribution, recomputes all support means, and refits the meta-analysis. The law of total variance can then be implemented by adding the average conditional covariance to the covariance of the bootstrap parameter estimates. Positive parameters may be sampled on a log scale. If the covariance of the external estimates is unavailable, as in the present motivating dataset, the meta-analytic standard errors remain conditional on the fitted support distribution and parameter perturbations should be reported as scenario-based sensitivity analyses rather than as unconditional confidence intervals. Near the boundary $\tau = 0$, the ordinary delta approximation is nonregular and a parametric bootstrap is preferred.

We also used decorrelated marginal residuals to assess model fit, following the general dose-response meta-analysis diagnostic principle described by Discacciati et al. 26. At fitted values, let $\hat{\mathbf{V}}_k = \boldsymbol{\Sigma}_k + \hat{\tau}^2 \mathbf{x}_k \mathbf{x}_k^\top$ and choose its Cholesky factor $\hat{\mathbf{L}}_k$ so that $\hat{\mathbf{V}}_k = \hat{\mathbf{L}}_k \hat{\mathbf{L}}_k^\top$. The decorrelated residual vector and generalized residual sum of squares are

$$\mathbf{z}_k = \hat{\mathbf{L}}_k^{-1}(\mathbf{y}_k - \hat{\beta}_1 \mathbf{x}_k), \quad Q_{\text{res}} = \sum_{k=1}^K \mathbf{z}_k^\top \mathbf{z}_k. \quad (2.18)$$

We report Q_{res} against the descriptive χ^2_{N-1} reference and the largest absolute decorrelated residual. Because τ , the working covariance, and in some applications the support distribution are estimated, this reference is a diagnostic rather than a formal goodness-of-fit test.

2.6. Sensitivity analysis

To assess the sensitivity of the proposed model, analyses were conducted under alternative assumptions for both the within-study correlation structure and the exposure distance distribution. Specifically, analyses were repeated across $\rho = 0, 0.25, 0.5, 0.75$, and 0.9 , as introduced previously in Section 2.1, as well as under several Gamma distribution specifications for residential distance. These included study-region-specific parameter estimates, an equal-weight pooled Gamma specification based on the four actual study-region parameter sets, an Iowa-augmented five-region external scenario, corresponding $\pm 20\%$ scale adjustments, and a source-quality scenario in which the pooled parameters from the North

Carolina and Wisconsin permitted-facility layers replaced the Pennsylvania producer-directory and Netherlands agricultural-business proxy parameters. Detailed settings and results are presented in the Results section below.

For a prespecified collection Ψ of support mappings, we summarize distributional sensitivity by the range $\{\hat{\beta}_1(\psi) : \psi \in \Psi\}$ and by the envelope formed from the smallest lower and largest upper confidence limits over Ψ . This is a descriptive sensitivity envelope, not a formal identified set or a simultaneous confidence interval. Without restrictions on the within-support distributions, nonparametric bounds can be uninformative: conditional means may approach their support endpoints and some resulting design contrasts may approach zero. Formal partial identification therefore requires a scientifically defensible admissible class and uniformly valid inference; the finite scenario set here is intended to make the dependence on working choices transparent.

2.7. Exposure distribution and distance-data construction

Exposure distribution. When modeling the association between residential proximity to animal feeding operations (AFOs) and lower respiratory disease (LRD) risk, it is important to first characterize the distribution of residential distance D to the nearest AFO. Here, D represents the target distance between a residence and the nearest AFO, where residences are those of community members and exclude residences of owners or workers at AFOs. In the illustrative first-stage construction, permitted-facility data were available for North Carolina and Wisconsin, whereas the Pennsylvania agricultural-producer directory and Netherlands livestock-classified agricultural-business layer were used as geospatial proxies. Accordingly, the latter fitted distributions represent distance to a recorded proxy site rather than a verified nearest-AFO distribution.

A key modeling challenge is that the exposure variable D is not directly reported in a continuous form across primary studies. Most observed odds ratios (ORs) are based on interval comparisons (e.g., within 1 mile vs. beyond 3 miles), while some compare model-based estimates at exact distances. A working continuous distribution for D is required to compute representative means for the positive-width intervals; zero-width intervals are evaluated directly at their exact point values and do not depend on this distribution.

Residential distances to the nearest AFO are expected to follow several general patterns, although the specific distribution may differ across geographic regions. Very small residential distances may be relatively uncommon because of land-use patterns, siting practices, and, in some jurisdictions, zoning or regulatory restrictions on residential development near AFOs. However, these factors are not assumed to operate uniformly across all countries or regions represented in the included studies. Extremely large distances may also be relatively uncommon because the study populations were drawn from regions with active AFO production. As a result, many residences may fall within a moderate distance range from the nearest AFO.

Consequently, the distribution of D is not expected to increase monotonically with distance. Instead, the number of residences may initially increase from very small distances, reach a maximum within a moderate distance range, and then gradually decline as distance increases. Although multiple AFOs may exist within a region, so that a residence may be far from some AFOs while remaining close to another, some residences may also be relatively far from all recorded facilities or proxy sites. The frequency and location of such residences depend on the settlement patterns, AFO density, and geographic characteristics of each study region.

The available regional distance-to-recorded-site data showed positive, right-skewed, and approximately unimodal distributions. Under these considerations, the working distribution of residential distance was represented using a Gamma distribution, which accommodates positive support, right-skewness, and a range of unimodal shapes. The parameters were estimated separately from the available

regional data, providing a flexible working representation of the residential distance distribution rather than imposing the same distance pattern across all study settings.

Formally, we model the residential distance to the nearest AFO as

$$D \sim \text{Gamma}(\alpha, \theta), \quad f(d) = \frac{1}{\Gamma(\alpha)\theta^\alpha} d^{\alpha-1} e^{-d/\theta}, \quad d \in [0, \infty). \quad (2.19)$$

The Gamma distribution is used as a flexible working model for the distribution of residential distance to the nearest AFO or its recorded-site proxy. Because the studies included in the meta-analysis span multiple geographic regions, the true distance distribution may vary across settings. Therefore, the meta-analysis uses study-region-specific Gamma parameters to reflect the residential distance structures represented in the included studies. Additional pooled, scale-adjusted, and registry-source Gamma distribution specifications were further evaluated through sensitivity analyses to assess the model under alternative assumptions about the residential distance distribution.

Estimation of α and θ . Regional residential and animal-facility or agricultural-site location data were assembled from public sources for the Netherlands, North Carolina, Pennsylvania, Wisconsin, and an Iowa reference region. Within each region, the population-weighted distribution of distance from a residential unit to its nearest recorded facility or proxy site was fitted by maximum likelihood. Study-region-specific Gamma estimates were primary. The four regions represented by included studies defined the pooled and $\pm 20\%$ pooled-scale sensitivities; Iowa was retained only in a separate five-region external scenario. Figure 2 shows the fitted distributions. The public source files, facility classifications, spatial processing, Netherlands data limitation, and pooled-parameter construction are documented in Supplementary Methods S1. The source-quality sensitivity replaces the two proxy-layer parameter sets with a pooled working distribution derived from the North Carolina and Wisconsin permitted-facility layers; it tests first-stage source choice rather than claiming geographic transportability of those registries.

2.8. Data sources and inclusion criteria

Because the proposed model defines exposure contrasts in terms of distance intervals to the nearest AFO, it is necessary to standardize all exposure measures into a common distance-based framework. In particular, each observation must correspond to a well-defined comparison between two distance intervals. The intervals may have positive or zero width; a reported exact distance is represented as a zero-width interval. Exposure measures that could not be mapped to a specific distance threshold or exact value were excluded, as they are not compatible with this support-based structure. For example, metrics such as odor annoyance levels or the number of AFOs within administrative areas do not provide a clear spatial threshold for defining exposure 28, 29. Similarly, non-distance-based exposure measures, such as interquartile range increases in PM_{10} or NH_3 , were excluded because they cannot be expressed within the distance-based framework 30.

Within the LRD outcome category of the living systematic review, 16 studies reported 37 distinct disease-specific outcomes through November 3, 2025. These studies were conducted across 16 locations in Croatia, Germany, the Netherlands, and the United States. Because some studies reported multiple effect estimates using different outcome definitions or subpopulations, we prioritized asthma-related outcomes, including asthma hospitalization, physician-diagnosed asthma, and asthma-related mortality, to identify comparable measurements across studies. When an included study did not report an asthma-related outcome, we retained an alternative LRD outcome, including pneumonia or mortality attributed to chronic lower respiratory disease.

The primary AFO working analysis retained all 33 mapped contrasts from eight studies. For example, “within 3 miles” versus “more than 3 miles” was represented as $(0, 3]$ versus $(3, d_{\max}]$, where $d_{\max}$ is the finite operational bound recorded in the curated extraction; it is not necessarily an observed source-study participant maximum. A right-unbounded $(3, \infty)$ sensitivity was also evaluated under the Gamma

working distribution. In the workbook and code, Type 1 denotes an explicit support pair. Type 2 denotes a binary-buffer proxy: the stated radius r is mapped to $(0, r]$ versus $(r, d_{\max}]$. Such presence indicators and animal-count measures are not assumed to be exact nearest-facility distances. They are retained as documented working mappings in the primary analysis and removed in the explicit-support restriction. In contrast, Schultz et al. 31 reported five restricted-cubic-spline comparisons at specific distances: 1, 1.5, 2, 2.5, and 3 miles, each compared with 5 miles. These comparisons were represented as zero-width intervals, for example $\{1.5\}$ versus $\{5\}$, so that the associated exposure contrast is $1.5 - 5 = -3.5$ miles and no integration is required. All distance measurements were converted to miles for consistency. Here, a zero-width support represents a model-evaluated spline contrast at the stated distances; it does not imply that all contributing participants were observed at those exact distances.

Animal type was not used to estimate animal-type-specific effects. Explicit species-specific distance intervals, including the Freidl poultry- and goat- facility categories, remain Type 1 supports and are eligible for the strict explicit-support restriction. Species-specific binary-buffer or presence indicators remain Type 2 proxy mappings.

In total, 33 effect estimates were drawn from eight studies 31–38, covering six distinct LRD outcomes. The primary working analysis contained 29 reported ORs and four hazard-ratio animal-count proxies from Simões et al. 38, analyzed as approximate ORs on the log scale. An OR-only restriction excluded the four HR records, and a strict explicit-support OR restriction excluded the binary-buffer and animal-count proxy mappings, retaining 15 ORs from four studies. This hierarchy does not imply epidemiologic interchangeability of the effect measures or exposure constructs. All 56 contrasts in the independent radon illustration were reported adjusted ORs. The AFO estimates were derived from different study designs and outcome definitions and represented prevalent disease, hospitalization, pneumonia, or mortality. For each observation, we retained the effect estimate and 95% confidence limits as extracted from the source article. As is common in meta-analysis based on published results, the corresponding individual participant data and group-level aggregate data, including exposure-group frequencies, event counts, and denominators, were generally unavailable. This dataset was used to demonstrate the proposed method rather than to provide a definitive substantive synthesis; revisions to the living review or its extraction may change the empirical estimates without changing the method.

2.9. Independent nonuniform-exposure illustration

To examine portability beyond residential distance under a source-supported nonuniform exposure distribution, we reconstructed the published residential radon–lung-cancer meta-analysis of Pavia et al. 39. Its web Table 2 reports adjusted ORs and 95% confidence intervals for study-specific categories of residential radon. The source analysis represented categories by a reported mean or median when available, otherwise by the midpoint, and chose proportional scores for open-ended categories before fitting study-level log-linear trends. These conventional point assignments make the dataset a direct illustration of the support-representation problem addressed here.

We retained the 16 studies reporting time-weighted mean residential radon in the common unit Bq/m^3 , yielding 56 non-reference adjusted OR contrasts. The Létourneau et al. study was excluded because the source table reports only cumulative exposure, which is not commensurate with time-weighted Bq/m^3 . Many categories were reported as rounded integers. We used contiguous half-unit cutpoints for those categories; for example, 74–147 was represented as $(73.5, 147.5]$, while explicitly reported decimal cutpoints were retained. This is a rounding/discretization convention. Only changing whether a single endpoint is included, not moving an endpoint, has no effect under a continuous distribution.

The disease meta-analysis does not provide the group frequencies needed to estimate within-support exposure distributions. We therefore used independent, outcome-free national residential-radon surveys. The primary first-stage model used approximately contemporaneous country-specific arithmetic means, geometric means, and geometric standard deviations from UNSCEAR 2000 and the WHO handbook 40,

41. Indoor radon is positive and right-skewed, and these sources summarize national concentrations using Lognormal parameters (Supplementary Table S1). For Sweden, whose historical table reports an arithmetic and geometric mean but no geometric standard deviation, we used the implied Lognormal value $\exp[\{2 \log(108/56)\}^{1/2}] = 3.14591897$ (3.146 to three decimals). A sensitivity analysis used the updated country-specific UNSCEAR 2024 parameters 42.

For every comparison and reference interval we calculated its Lognormal conditional mean and defined x_{kj} as their difference. This first-stage calculation used no exposure-category frequencies, case/control counts, or outcomes. The contrast-level model used only those representatives, the adjusted ORs, and their confidence limits. We used $\rho = 0.5$ as the primary exchangeable within-study working correlation and repeated the analysis at $\rho = 0, 0.25, 0.5, 0.75$, and 0.9 . To show the ambiguity of conventional scoring, we also used interval midpoints with each right-open category assigned 1.2, 1.5, or 2.0 times its lower bound. These are sensitivity rules rather than claims about the source authors' exact proportional scores.

3. Results

3.1. Distance Density Function

As shown in Figure 2, the estimated Gamma working distributions differed across the geographic regions included in the analysis, with the Netherlands curve appearing more concentrated than the U.S. regional curves. The regional curves varied in spread and tail behavior, indicating that the assumed residential distance distribution was not identical across settings. The Netherlands fitted mean was approximately 0.58 miles, but this value reflects a livestock-keyword-filtered agricultural-business layer in one municipality, Eijdsden-Margraten, together with its limited geographic coverage. It should not be interpreted as an estimate of national residential distance to a verified AFO.

The figure also includes the Iowa-augmented five-region pooled Gamma specification as an external scenario. The pooled specification used for the principal sensitivity analyses was instead constructed from the four regions represented by included studies. Overall, the regional variation shown in Figure 2 supports the use of study-region-specific Gamma parameters and motivates the additional pooled specifications.

3.2. AFO motivating stress-test illustration

Study structure and working analysis hierarchy. The primary AFO working analysis comprised 33 analytic log-OR-scale observations from eight studies: 29 reported ORs and four hazard ratios treated as approximate ORs. It retained the documented Type 2 binary-buffer and animal-count support mappings because heterogeneous and indirect supports are the motivating data problem. Because these choices add support-mapping and effect-measure assumptions, we also conducted an OR-only restriction excluding the four hazard ratios and a strict explicit-support OR restriction excluding Type 2 mappings. The strict restriction contains 15 ORs from four studies, including positive-width categories and the five restricted-cubic-spline comparisons from Schultz et al. 31, represented as contrasts between zero-width intervals at the reported exact distances.

For every retained observation, x_{kj} was the difference between the mapped comparison and reference supports in Equation (2.9). Where a source category was genuinely open-ended, the working analysis used the finite operational bound recorded in the curated dataset. Evaluating genuinely open-ended categories in the strict restriction as (d, ∞) under the Gamma working distribution gave the same rounded estimates.

Primary eight-study working analysis. Under the study-region-specific Gamma distributions and working correlation $\rho = 0.5$, the random-effects estimate was $\hat{\beta}_1 = -0.069$ (standard error 0.051; model-based 95% confidence interval -0.170 to 0.031), with $\hat{\tau} = 0.129$. The study-count t_7 interval was

−0.191 to 0.052; the 2,000-replicate studentized parametric-bootstrap interval was −0.198 to 0.054 and is reported only as an additional model-based sensitivity. Midpoint scoring of all 33 mapped contrasts gave $\hat{\beta}_1 = -0.034$ (95% confidence interval −0.085 to 0.017), with $\hat{\tau} = 0.070$.

Restriction analyses. The OR-only restriction contained 29 comparisons from seven studies and gave $\hat{\beta}_1 = -0.064$ (95% confidence interval −0.185 to 0.057), with $\hat{\tau} = 0.146$. The strict explicit-support OR restriction contained 15 comparisons from four studies and gave $\hat{\beta}_1 = -0.137$ (standard error 0.103; 95% confidence interval −0.339 to 0.064), with $\hat{\tau} = 0.194$. Its study-count t_3 interval was −0.464 to 0.189. These wide intervals reflect that only four independent studies enter the strict restriction. Midpoint scoring of those 15 records gave $\hat{\beta}_1 = -0.092$ (95% confidence interval −0.217 to 0.033). Table 1 displays the hierarchy.

For the primary working analysis, the 30 combinations of working correlation and Gamma distribution specification gave slopes from −0.084 to −0.035 and a descriptive model-based interval envelope of (−0.193, 0.039). A one-most-precise-contrast-per-study analysis gave −0.047 (95% confidence interval −0.145 to 0.051). Leave-one-study-out results, study information weights, covariance condition numbers, the mapping-fragility diagnostic, and the full support-mapping audit are reported in the supplement. We did not rely on CR2 inference for the AFO illustration because the primary analysis has only eight independent clusters and the strict restriction has four; the study-count- t , correlation-grid, and one-contrast-per-study results are reported as transparent finite-study sensitivities. Across these analyses, the direction was negative and uncertainty included zero, but the slope magnitude depended on support and proxy choices.

3.3. Principal nonuniform-exposure illustration

The radon dataset comprised 56 adjusted OR contrasts from 16 studies. Under the primary historical country-specific Lognormal distributions and $\rho = 0.5$, the random-effects slope was $\hat{\beta}_1 = 0.000781$ per Bq/m³ (standard error 0.000239; 95% confidence interval 0.000314 to 0.001249). This corresponds to an OR of 1.081 per 100 Bq/m³ (95% confidence interval 1.032 to 1.133), or 1.124 at 150 Bq/m³ relative to zero (95% confidence interval 1.048 to 1.206). The estimated random-slope heterogeneity was $\hat{\tau} = 0.000468$ per Bq/m³, with a profile restricted-likelihood interval from zero to 0.001219.

Pavia et al.'s original two-stage analysis reported an OR of 1.24 at 150 Bq/m³ (95% confidence interval 1.11 to 1.38) 39. On its reported log-linear scale, this corresponds to approximately 1.154 per 100 Bq/m³. Our estimate of 1.081 per 100 Bq/m³ is numerically close to the individual-participant-data European pooled estimate for measured radon: Darby et al. reported an 8.4% increase (95% confidence interval 3.0% to 15.8%), with the detailed analysis reporting an excess relative risk of 0.08 (0.03 to 0.16) per 100 Bq/m³ 43, 44. The North American individual-participant-data pool reported an OR of 1.11 (1.00 to 1.28) per 100 Bq/m³ 45. The estimates should not be expected to coincide: their analysis first assigned means, medians, midpoints, or proportional open-tail scores and estimated a separate trend within each study, whereas ours analyzes the adjusted contrasts jointly under a working covariance and an explicit support distribution. The comparison is therefore methodological rather than a claim that the source analysis estimated the same model incorrectly.

The agreement with the pooled individual-data estimates is a contextual, non-independent concordance check rather than external validation: the pooled analyses reuse several primary studies represented in Pavia et al. and apply different participant-level adjustment and exposure models. The appropriate European comparator is the 8.4% measured-radon coefficient. Darby et al.'s 16% estimate corrects measured radon for regression dilution and therefore targets usual radon, whereas neither the Pavia aggregate table nor the present analysis applies that measurement-error correction.

The result was stable to the independent distribution source. Using the country-specific UNSCEAR 2024 parameters gave an OR per 100 Bq/m³ of 1.083 (95% confidence interval 1.027 to 1.143). Across $\rho = 0, 0.25, 0.5, 0.75$, and 0.9, the historical-distribution ORs per 100 Bq/m³ were 1.109, 1.091,

1.081, 1.066, and 1.035, respectively. The confidence interval included one only at the deliberately extreme value $\rho = 0.9$, showing that the working covariance can materially affect inference. The 15-degree-of-freedom t interval for the primary slope was 0.000273 to 0.001290, and the studentized parametric-bootstrap interval was 0.000212 to 0.001325.

A CR2 cluster-sandwich variance sensitivity after whitening by the fitted marginal covariance, paired with a $t_{K-1} = t_{15}$ critical value, gave standard error 0.000237 and an OR of 1.081 per 100 Bq/m³ (95% confidence interval 1.028 to 1.137). This was not a Satterthwaite degrees-of-freedom correction. Retaining only the most precise contrast from each study gave an OR of 1.163 (study-count- t interval 1.064 to 1.271) and $\hat{\tau} = 0$. The former closely agrees with the model-based result; the latter shows that discarding within-study contrast information can materially alter the slope. Study information weights and covariance condition numbers are reported in the supplement.

In the leave-one-study-out radon analysis, the OR per 100 Bq/m³ ranged from 1.072 (omitting Wang et al.) to 1.094 (omitting Darby et al.); all 16 confidence intervals excluded one. The primary radon result was therefore not driven by a single published study. The individual refits are reported in Supplementary Table S4.

In contrast, the conventional midpoint analysis required an additional rule for every right-open category. Assigning 1.2, 1.5, or 2.0 times the lower bound gave ORs per 100 Bq/m³ of 1.102, 1.080, and 1.057, respectively (Table 2). This variation is created solely by the analyst-chosen tail score. The Lognormal mapping instead supplies finite conditional means for the open supports and makes the distributional assumption explicit and independently auditable. The radon result remains a methodological illustration rather than a new causal risk assessment.

3.4. Simulation study

The simulation retained the empirical mixed-support design: 28 observations involving positive-width intervals and five Schultz observations comparing zero-width intervals at exact distances. Thus, the same simulation evaluated both support widths in a common analysis. For each positive-width support, its representative value was computed under the setting-specific distribution; for each zero-width support, it was the exact point distance.

Random-effects model Table 3 summarizes bias and RMSE across the 3×3 grid. Bias in $\hat{\beta}_1$ ranged from -0.002 to 0.001 . Its RMSE was approximately 0.028 when $\tau = 0.065$, 0.046–0.047 when $\tau = 0.129$, and 0.084–0.086 when $\tau = 0.259$. Bias in $\hat{\tau}$ ranged from -0.005 to -0.009 , with RMSE increasing from approximately 0.027 to 0.066–0.067 across the same heterogeneity levels. Thus, point-estimation error was driven mainly by the amount of between-study heterogeneity rather than the magnitude of the common slope.

Performance across study counts The experiment in Table 4 uses $K = 10, 20$, and 30 to examine generalizability. Absolute slope bias was below 0.001 at every study count and RMSE decreased from 0.045 to 0.026. Normal coverage was 0.915, 0.933, and 0.936; the corresponding study-count- t coverage was 0.948, 0.948, and 0.948. Under a null slope, Normal type-I error was 0.083, 0.064, and 0.056, compared with 0.053, 0.051, and 0.047 using the t critical value. Monte Carlo standard errors were at most 0.0040 for these cells. Thus, the t interval improved calibration but did not achieve nominal performance uniformly at small K ; it remains a sensitivity analysis.

Fixed-effect model Table 6 summarizes the fixed-effect GLS results under three slope values and no between-study heterogeneity. Absolute bias was less than 0.001, and RMSE was approximately 0.004 in all three settings.

Comparison to midpoint assignment Supplementary Tables S7 and S8 compare bias and RMSE. Under a true Uniform distribution, conditional means and midpoints produced identical results, as required, including identical treatment of the zero-width intervals. Bias in $\hat{\beta}_1$ was at most 0.001 in magnitude. Under a true Gamma distribution, the correctly specified conditional-mean approach again

had small bias: at most 0.001 for $\hat{\beta}_1$ and 0.010 for $\hat{\tau}$. Midpoint assignment produced $\hat{\beta}_1$ bias ranging from 0.025 to 0.107 and downward $\hat{\tau}$ bias ranging from -0.036 to -0.171 . This comparison shows that the difference between the methods concerns the positive-width intervals; both methods evaluate a zero-width interval at its exact point.

Stress tests Supplementary Table S9 reports the additional misspecification results at the empirical calibration values unless otherwise noted. Fitting a Gamma working distribution when the true distribution was a moment-matched Lognormal produced little additional bias. Bias also remained small when the true $d_{\max}$ values differed by 20% or the true within-study correlation was 0.3 or 0.7. Under individual-level logistic aggregation, bias in $\hat{\beta}_1$ ranged from 0.004 to 0.007 over the three slope magnitudes, consistent with the first-order calculation in Section 2.4. Scenario-based uncertainty in the regional Gamma parameters also had little effect: with 10% and 20% coefficients of variation for both shape and scale, $\hat{\beta}_1$ bias was < 0.002 in magnitude and RMSE was approximately 0.046. Under the nonlinear individual-level logistic models, $\hat{\beta}_1$ bias increased to 0.024 for $c = 0.05$ and 0.044 for $c = 0.10$, quantifying the interpretation cost of fitting a common linear slope when the true dose–response is curved; Normal coverage fell to 0.868 and 0.807, respectively, and study-count- t coverage to 0.905 and 0.861. In contrast, fitting the Gamma representative means when the true interval distribution was Uniform produced substantial bias: $\hat{\beta}_1$ bias was -0.290 with RMSE 0.399, and $\hat{\tau}$ bias was 0.728 with RMSE 0.800; Normal and t coverage were 0.823 and 0.904. This deliberately strong misspecification demonstrates that the working exposure distribution and linearity assumption are substantive modeling choices that should be supported by external information and examined through sensitivity analysis.

Summary of simulation findings Taken together, the simulations show small point-estimation bias under correct specification in the observed mixture of positive- and zero-width supports, little additional bias under moderate Gamma-parameter uncertainty, modest approximation bias under individual-level logistic aggregation, and increasing bias under curvature or severe exposure-distribution misspecification. The results support the proposed point estimator while also defining the practical role of exposure-distribution diagnostics and sensitivity analysis. Convergence was 100% in every reported setting; $\tau = 0$ boundary rates are reported in the supplement and were small except, as expected, in low-heterogeneity settings. Finite-study intervals are therefore presented as sensitivity analyses rather than as a separate claim of nominal small-sample calibration.

4. Discussion

In this study, we developed and evaluated a contrast-level aggregate-data dose–response meta-analysis method for heterogeneous exposure or intervention supports, including positive-width intervals and zero-width intervals representing exact values. The principal radon application used 56 adjusted OR contrasts from 16 studies and externally sourced nonuniform exposure distributions. The AFO application retained the difficult motivating data structure in an eight-study working analysis, with OR-only and explicit-support restrictions separating effect-measure and mapping assumptions. Both applications illustrate the method rather than provide new causal risk assessments.

The main contribution of this work is that it provides a structured way to conduct contrast-level aggregate-data dose–response meta-analysis when studies report heterogeneous exposure intervals and the corresponding individual participant and grouped exposure–outcome data are unavailable. More generally, the support-to-design framework includes intervals, points, and their mixtures. Conventional point-dose DRMA is not displaced: it is recovered exactly when all support widths are zero, conditional on the same basis, sampling covariance, and random-effects specification. No prior method we identified is applicable to the complete input combination studied here. Midpoint or endpoint substitution can make a conventional assigned-dose analysis executable, but only by adding a point-score approximation that is not contained in the reported data. By using the conditional mean distance within each positive-width interval and the reported distance itself for each zero-width interval, the framework maps published

adjusted contrasts to a common design without recoding all comparisons to a common reference support. The zero-width definition completes the framework through the shrinking-interval extension.

The method was motivated by a data configuration encountered in the systematic review to which the usual approaches were not applicable: the available observations were already-adjusted pairwise association estimates, their comparison and reference supports differed across observations, and the group-level data needed by interval-censored or dose-pre-analysis methods were not reported. Treating each interval by an assigned midpoint would have been a practical ad hoc approximation, but would not have made the underlying within-support assumption or its uncertainty explicit. The proposed method turns that step into a defined statistical mapping. Proposition 3 provides the main approximation justification: under an individual-level logit-linear model, the conditional-mean contrast recovers the first-order aggregate log-odds term, whereas midpoint scoring generally adds a first-order error. The supplementary projection result adds the secondary properties that the conditional mean is the minimum-mean-squared-error constant representation of exposure and is coherent under support aggregation, while Proposition 1 shows that the operator preserves support-averaged linear predictors. These properties, together with the ability to retain arbitrary reported support pairs and mix exact values with intervals, distinguish the framework from simply substituting one set of assigned category scores for another.

The empirical AFO analysis serves as a motivating stress test, not as a definitive substantive synthesis of LRD effects. Its primary working analysis retains all eligible mapped contrasts; the OR-only and strict explicit-support restrictions show the consequences of removing effect-measure and support-mapping assumptions. Consequently, the methodological contribution does not depend on those proxy choices or on the numerical result of this application. The choices affect the illustrative estimate, whereas the positive-width and zero-width support formulation remains applicable when the contrast dataset is updated or replaced.

The principal radon analysis demonstrates that the same support-to-design mapping is not specific to residential distance and need not rely on a Uniform distribution. It uses adjusted summary ORs and confidence intervals together with independently published country-level Lognormal exposure distributions; no exposure-group frequencies or outcome counts enter either the first-stage mapping or the contrast-level estimator. The similar results under historical and 2024 survey parameters show little sensitivity to those two national parameter sets but do not establish source-population distribution correctness, while the range of results under alternative proportional open-tail scores shows the otherwise unresolved analyst choice in conventional midpoint scoring. The same downstream GLS or REML estimator consequently handles interval-derived means, conventional point doses, and exact zero-width supports.

Figure 3 displays the direct benchmark against conventional scoring. In the AFO illustration, conditional-mean and midpoint support mappings yield visibly different slope estimates. In the radon illustration, changing only the analyst-selected score for right-open categories changes the pooled OR. The figure is descriptive rather than a ranking of working distributions; it shows why the first-stage representation must be reported and defended.

In practice, the working exposure distribution should be selected using a prespecified evidence hierarchy. Reported category means or exposure-group frequencies can identify support representatives directly; otherwise an independent distribution from the source population or a closely justified external population can be used. Plausible parametric families, tail treatment, and first-stage parameter uncertainty should then be varied. The outcome contrasts must not be used to choose the distribution that makes the synthesis most favorable. If no scientifically defensible distribution or admissible class is available, analysts should report midpoint or endpoint scoring only as an explicit approximation, present support-based sensitivity bounds where informative, or decline a quantitative dose-response synthesis.

The framework is also independent of study design: supports may arise from measured observational exposures or randomized intervention targets. Published meta-analyses of heterogeneous hemoglobin target ranges in chronic kidney disease trials and heterogeneous, sometimes overlapping, oxygen target

ranges in mechanically ventilated patients provide motivating clinical-trial data structures 14, 15. We cite these as examples of the support problem rather than as additional empirical analyses. In a trial application, the exposure variable would represent the randomized target strategy; achieved post-randomization values should not be substituted for the assigned target without changing the estimand.

Adding the four hazard-ratio/count proxy contrasts to the OR-only restriction did not materially change the direction of the random-effects slope. This sensitivity result does not establish that odds ratios and hazard ratios are interchangeable; instead, it shows that the methodological illustration and its qualitative random-effects interpretation were not dependent on retaining those four records. In a substantive review, effect measures should ordinarily be harmonized using an estimand-specific conversion supported by the required outcome information, or analyzed separately when such a conversion is not defensible.

Limitations Several data-related issues should be considered when interpreting these findings. The AFO data come from a living systematic review and may change as study selection and extraction are updated; such changes affect the numerical illustration but not the support-based formulation. The radon analysis is the principal empirical demonstration, but national surveys may not reproduce the exposure distributions of case-control source populations, particularly when studies sampled high-radon regions. We therefore used approximately contemporaneous parameters as primary, current parameters as a sensitivity, and varied open-tail scoring. The reported uncertainty is conditional on these external parameters and does not constitute a new causal radon-risk assessment. Exact within-study covariances were unavailable in both applications and were approximated using a working correlation; the radon CR2 and one-contrast-per-study analyses and the correlation grids quantify selected consequences. The primary AFO working analysis retains binary-buffer, animal-count, and hazard-ratio proxies under stated assumptions; the OR-only and explicit-support restrictions evaluate their removal and do not imply that these constructs or effect measures are epidemiologically interchangeable.

More generally, adjusted category ORs are noncollapsible and may differ across studies because of adjustment sets and standardization procedures. The unconditional support means used here need not equal the covariate-conditional or outcome-risk-weighted exposure means relevant to a particular adjusted coefficient. The method therefore estimates the stated support-level working slope; it does not remove confounding, harmonize incompatible adjusted estimands, or identify an individual-level causal effect.

Residential distance to the nearest AFO is also an imperfect proxy for actual exposure to AFO emissions. A distance-only metric does not account for wind direction or speed, atmospheric dispersion, facility size, animal population, emission intensity, topography, or exposure from multiple nearby operations. Consequently, two residences located the same distance from an AFO may experience substantially different exposures; for example, a residence located upwind may have lower airborne exposure than one located the same distance downwind. The estimated slope should therefore be interpreted as an association with residential proximity rather than as the causal effect of exposure to AFO emissions. In addition, the regional geospatial data used to estimate distance distributions were not equally available across settings. The North Carolina and Wisconsin sources were permitted-facility layers. The Pennsylvania source was a public agricultural-producer directory rather than the official state CAFO permit layer, and the Netherlands source described agricultural businesses in Eijsden-Margraten that were filtered by livestock keywords rather than verified AFO permits. These two fitted distributions are therefore first-stage geospatial proxies. Their update timing, classification, and geographic coverage may not match the source populations of the association studies. The source-quality sensitivity replaced both proxy parameter sets with a pooled North Carolina/Wisconsin permitted-facility working distribution; it changed the magnitude but not the negative, null-including random-effects conclusion. The original regional fitting routines requested numerical Hessians, but the observation-level distance inputs and fitted covariance matrices were not retained in the reproducibility bundle, so a defensible covariance cannot be reconstructed from the stored point estimates. The reported standard

errors are therefore conditional on the first-stage parameters. The scenario analyses assess broader source, proxy, and parameter sensitivity but are not unconditional confidence intervals; Equation (2.17) and the accompanying bootstrap give the appropriate propagation procedure when a first-stage covariance estimate is retained. A fitted Hessian alone would not capture the structural transportability uncertainty of choosing a proxy layer. Interpretation is also limited by the design and reporting of the primary studies. Previous evaluations of this evidence base have identified concerns involving confounder selection, possible adjustment for mediators or colliders, and the population assumptions required for causal interpretation of prevalence-based effect measures 16, 17.

Future Directions The current framework also has several possible extensions. In this analysis, a single common slope structure was used for all included distance-related comparisons. Future work could include study-level covariates or other effect modifiers to help explain sources of between-study heterogeneity. Although the present method is formulated for contrast-level aggregate data, the underlying interval model could be extended to richer data structures. When group-level aggregate data are available as event counts and denominators within exposure intervals, the interval-specific outcome probabilities could be modeled directly using a binomial likelihood:

$$R_{kA} \sim \text{Binomial}(N_{kA}, p_{kA}), \quad \text{logit}(p_{kA}) = \gamma_k + (\beta_1 + b_{1k})\mu_k(A).$$

In this formulation, the study-specific baseline parameter γ_k would be estimable because absolute event rates are available. However, adjustment for participant-level covariates would require additional reported information or individual participant data.

The model could also be extended to individual participant data. One approach would summarize the participant-level data within each study into interval-specific contrasts and their covariance matrix and then apply the present contrast-level aggregate-data method. Alternatively, a one-stage model could use participant-level outcomes directly with study-specific baseline terms and fixed or random exposure slopes. Development and evaluation of these group-level aggregate-data and individual-participant-data extensions are beyond the scope of the present study. Future work could also incorporate additional studies or updated effect estimates as the living systematic review expands. A larger summary-level dataset would allow more precise estimation of between-study heterogeneity and provide greater ability to evaluate whether the negative distance–response pattern observed here represents a consistent association across studies.

Finally, the same support-based framework could be applied to other extracted outcomes and exposure categories from the living systematic review, including upper respiratory diseases, antimicrobial resistance, infectious diseases, and gastrointestinal diseases. The framework could also be expanded to evaluate animal-type-specific or production-category-specific exposure definitions when sufficient data are available, which may help determine whether distance-related associations differ by AFO type.

Conclusion Overall, the proposed model is a support-based extension of conventional point-dose dose–response meta-analysis for contrast-level aggregate data with positive-width intervals or mixtures of intervals and points. It contains conventional contrast-design DRMA as an exact special case when all widths are zero under the same basis, covariance, and random-effects specification. No prior method we identified is applicable to the complete combination of adjusted arbitrary-pair contrasts, mixed point and interval supports, within-study dependence, and absence of grouped frequencies or outcome counts. Midpoint or endpoint assignment can be used only by adding an approximation whose within-support assumptions must be justified. The environmental mixed-support application and the radon application with independently sourced exposure-distribution information demonstrate the same support-to-design mapping under different first-stage information; published randomized-trial target-range examples show that the underlying support problem also occurs in clinical research. The principal contribution is the general support-based model rather than any substantive estimate.

5. Figures

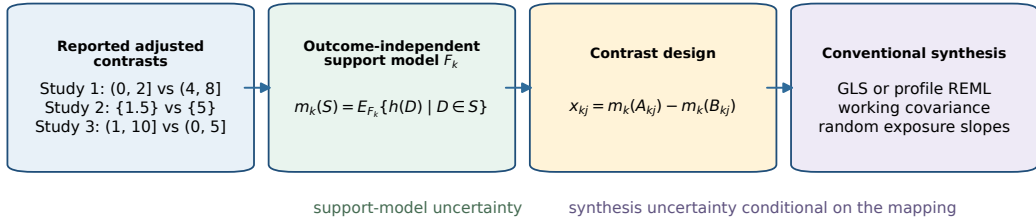


Figure 1. Support-to-design framework. Each reported adjusted contrast retains its observation-specific comparison and reference supports. An outcome-independent exposure model maps each support to an average dose-response basis vector; their difference forms the contrast design passed to conventional synthesis. Support uncertainty and synthesis uncertainty are distinct..

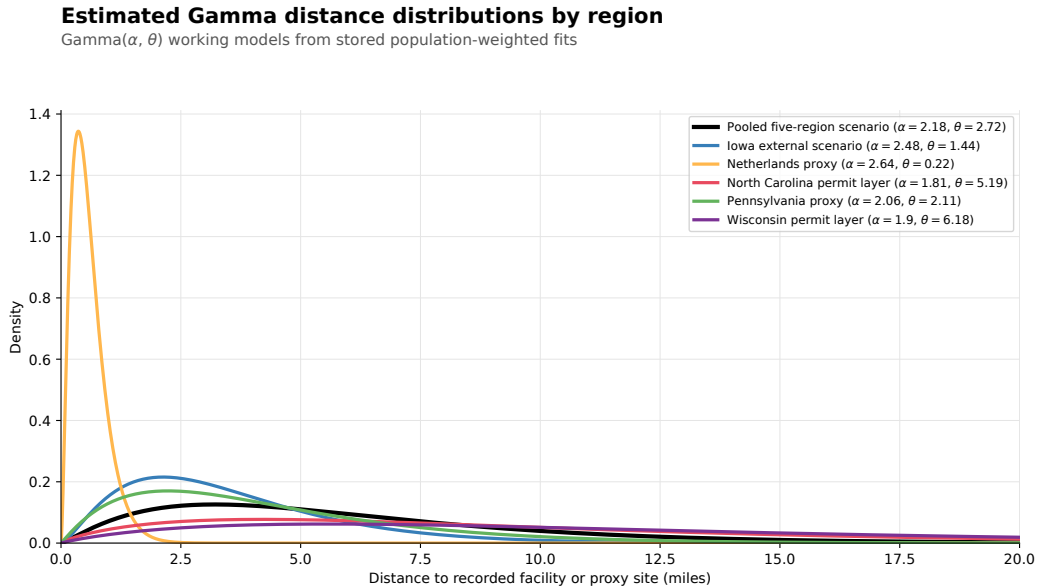


Figure 2. Illustrative Gamma working distributions for residential distance to recorded facilities or proxy sites, based on population-weighted maximum likelihood. North Carolina and Wisconsin use permitted-facility layers; Pennsylvania uses an agricultural-producer directory and the Netherlands uses livestock-classified agricultural businesses in one municipality. Iowa and the displayed five-region pooled curve are external scenarios; the primary pooled sensitivities use only the four regions represented by included studies..

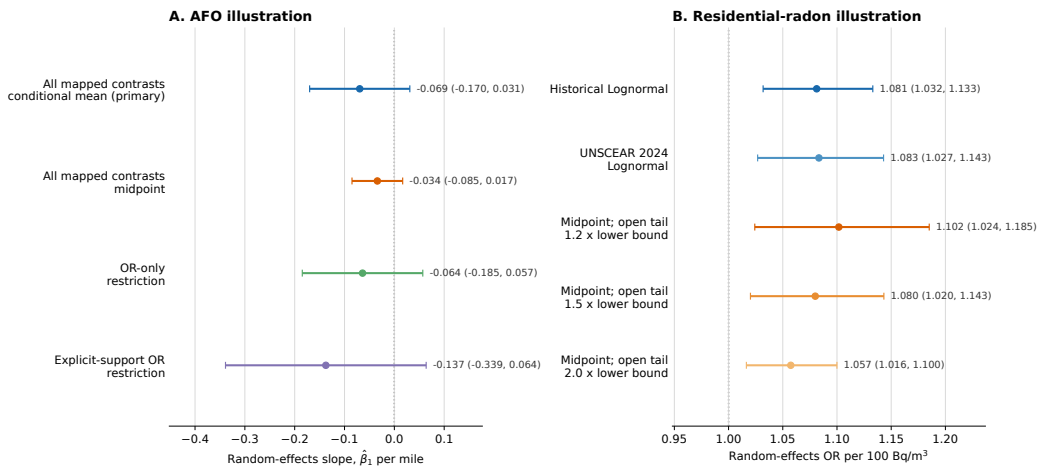


Figure 3. Direct comparison of support-representation assumptions. Panel A compares the primary eight-study AFO analysis under the regional Gamma conditional-mean and midpoint mappings, followed by the OR-only and strict explicit-support restrictions. Panel B compares the residential-radon random-effects OR per 100 Bq/m³ under historical and current country-specific Lognormal mappings and under midpoint scoring with three analyst-selected rules for right-open categories. Horizontal lines are model-based 95% confidence intervals; the vertical dotted line in Panel B marks the null OR of 1..

6. Tables

Table 1. Primary and restriction analyses for the AFO illustration under random effects and working correlation $\rho = 0.5$.

Analysis	Support representation	<i>N</i>	<i>K</i>	$\hat{\beta}_1$	SE	95% CI	$\hat{\tau}$
All mapped contrasts (primary)	Conditional mean/point	33	8	-0.069	0.051	(-0.170, 0.031)	0.129
All mapped contrasts	Midpoint/point	33	8	-0.034	0.026	(-0.085, 0.017)	0.070
OR-only restriction	Conditional mean/point	29	7	-0.064	0.062	(-0.185, 0.057)	0.146
Explicit-support OR restriction	Conditional mean/point	15	4	-0.137	0.103	(-0.339, 0.064)	0.194

N is the number of contrasts and *K* the number of independent studies. Confidence intervals are model-based Normal intervals. The primary study-count- t_7 interval was (-0.191, 0.052) and the studentized parametric-bootstrap sensitivity interval was (-0.198, 0.054). For the explicit-support restriction, the t_3 interval was (-0.464, 0.189). Restricted-likelihood values are not compared across exposure-design mappings.

Table 2. Principal residential-radon-lung-cancer illustration from adjusted summary contrasts and external Lognormal exposure distributions..

Exposure representation	ρ	$\hat{\beta}_1$ per Bq/m ³	OR per 100 Bq/m ³ (95% CI)	$\hat{\tau}$
Historical Lognormal	0	0.001031	1.109 (1.045, 1.176)	0.000809
Historical Lognormal	0.25	0.000872	1.091 (1.040, 1.145)	0.000478
Historical Lognormal (primary)	0.5	0.000781	1.081 (1.032, 1.133)	0.000468
Historical Lognormal	0.75	0.000643	1.066 (1.014, 1.122)	0.000674
Historical Lognormal	0.9	0.000342	1.035 (0.971, 1.103)	0.001090
UNSCEAR 2024 Lognormal	0.5	0.000800	1.083 (1.027, 1.143)	0.000598
Midpoint; open upper 1.2 <i>a</i>	0.5	0.000968	1.102 (1.024, 1.185)	0.000887
Midpoint; open upper 1.5 <i>a</i>	0.5	0.000769	1.080 (1.020, 1.143)	0.000668
Midpoint; open upper 2.0 <i>a</i>	0.5	0.000558	1.057 (1.016, 1.100)	0.000423

The analysis contains 56 non-reference adjusted OR contrasts from 16 studies reporting time-weighted residential radon in Bq/m³. The historical primary distributions use approximately contemporaneous national survey parameters; UNSCEAR 2024 is a distribution-source sensitivity. In the point-score rows, *a* is the lower bound of a right-open support. No exposure- group frequencies, event counts, or denominators are used. For the primary analysis, the study-count t interval for $\hat{\beta}_1$ was (0.000273, 0.001290); the CR2 sensitivity gave OR 1.081 per 100 Bq/m³ (95% CI 1.028–1.137) using the CR2 variance and a t_{15} critical value, without a Satterthwaite degrees-of-freedom approximation. The studentized parametric-bootstrap sensitivity interval was (0.000212, 0.001325).

Full random-effects and fixed-effect sensitivity grids are reported in Supplementary Tables S5 and S6. Detailed midpoint-comparison and stress-test results are reported in Supplementary Tables S7–S9.

Table 3. Bias and root mean squared error (RMSE) from the mixed-support 3×3 profile REML simulation grid. Each setting used $R = 5,000$ replicates..

$\beta_{1,\text{true}}$	τ_{true}	Bias (RMSE), $\hat{\beta}_1$	Bias (RMSE), $\hat{\tau}$
-0.035	0.065	0.001 (0.028)	-0.006 (0.027)
-0.069	0.065	< 0.001 (0.028)	-0.005 (0.027)
-0.139	0.065	> -0.001 (0.028)	-0.006 (0.027)
-0.035	0.129	> -0.001 (0.046)	-0.005 (0.039)
-0.069	0.129	< 0.001 (0.046)	-0.006 (0.039)
-0.139	0.129	< 0.001 (0.047)	-0.005 (0.039)
-0.035	0.259	-0.002 (0.084)	-0.008 (0.067)
-0.069	0.259	> -0.001 (0.086)	-0.009 (0.066)
-0.139	0.259	-0.001 (0.086)	-0.007 (0.066)

Values smaller than 0.0005 in magnitude are displayed as < 0.001 or > -0.001 according to direction.

Table 4. Point and interval performance across numbers of independent study blocks under the empirical slope and heterogeneity. Each setting used 25 resampled empirical support-and-covariance templates and $R = 5,000$ replicates..

K	Bias	RMSE	Normal cov.	t cov.	Normal width	t width	Normal type I	t type I	$\tau = 0$ rate
10	0.0001	0.0451	0.9150	0.9482	0.1710	0.1974	0.0832	0.0526	0.0020
20	0.0007	0.0325	0.9326	0.9482	0.1243	0.1328	0.0640	0.0508	0
30	-0.0007	0.0263	0.9356	0.9482	0.1021	0.1065	0.0556	0.0470	0

The data-generating values were $\beta_{1,\text{true}} = -0.069$ and $\tau_{\text{true}} = 0.129$. Type-I error was evaluated in separate null-slope cells at the same K and τ . Monte Carlo standard errors for coverage and type-I error were at most 0.0040; all fits converged. The identical 0.9482 entries at $K = 10, 20$, and 30 were reproduced from simulations with distinct fixed seeds.

Table 5. Selected exposure-distribution and nonlinear stress tests at $K = 10$ and $R = 5,000$..

Scenario	$\hat{\beta}_1$ bias	RMSE	$\hat{\tau}$ bias	RMSE	Normal cov.	t cov.
Correct Gamma working model	0.0006	0.0462	-0.0066	0.0393	0.9066	0.9376
Uniform true; Gamma fitted	-0.2898	0.3994	0.7283	0.8000	0.8232	0.9038
Nonlinear logit, $c = 0.05$	0.0242	0.0516	-0.0112	0.0397	0.8684	0.9054
Nonlinear logit, $c = 0.10$	0.0436	0.0626	-0.0084	0.0385	0.8066	0.8610

The true slope and heterogeneity were -0.069469 and 0.129253. Monte Carlo standard errors for coverage were at most 0.0056; all fits converged. Full stress-test results and boundary rates are in the supplement.

Table 6. Point-estimation performance of the fixed-effect GLS estimator. Each setting used $R = 5,000$ mixed-support replicates..

$\beta_{1,\text{true}}$	Mean $\hat{\beta}_1$	Bias	RMSE
-0.005	-0.005	> -0.001	0.004
-0.011	-0.011	< 0.001	0.004
-0.022	-0.022	< 0.001	0.004

Bias values smaller than 0.001 in magnitude are displayed as < 0.001 or > -0.001 according to direction.

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Use of Artificial Intelligence Tools. OpenAI ChatGPT and the OpenAI Codex desktop application were accessed during manuscript development through August 16, 2026, through the interfaces available from <https://openai.com/>. Model/version labels were not consistently retained, so no specific model number is asserted. The tools assisted with language editing, literature-search planning, code review and generation, implementation of author-specified simulation designs, and interpretation checks. They did not create or alter the source data and did not determine the statistical estimand or conclusions. The authors independently checked cited sources, executed every analysis, regenerated simulation outputs from the versioned code, reviewed the resulting text, and take full responsibility for the manuscript, transformations, conclusions, and code.

Data Availability Statement. The analysis-ready contrast workbooks, complete second-stage application and simulation code, validation tests, seeds, machine-readable results, and reproducibility instructions are available at <https://github.com/ChongWangStat/IntervalDRMA>; a permanent versioned archive DOI will be inserted before submission. The radon support and outcome data are reconstructed from the cited public reports. The AFO association estimates come from the living systematic-review dataset and co-author-curated materials. Public geospatial sources underlie the stored regional AFO working- distribution parameters, but the original observation-level distance files, fitting scripts, and fitted covariance matrices were not retained in the archive. Consequently, the second-stage AFO analyses are reproducible from the stored parameters, whereas the historical first-stage fits are not fully reconstructible. North Carolina and Wisconsin use permitted-facility sources; Pennsylvania and the Netherlands use agricultural-site proxies as documented in Supplementary Methods S1.

Ethical Standards. This study used published aggregate effect estimates and public geospatial data and involved no new collection of individual participant or animal data. Institutional ethics approval was therefore not required.

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